On the discussion questions regarding Frege's preface to Begriffsschrift and Russell's chapter 2 from Our Knowledge of the External World (September 9, 2025)

Here are a couple of thoughts and observations on the discussion questions. Comments are welcome!

All groups:
"Are all philosophical problems amenable to rational (logical) treatment? If so, why? If not, what would be examples of such problems? And what would be their characteristics?"

If one takes a broad look at modern analytic philosophy, as it developed since the beginning of the twentieth century to the present-day, one can definitely discern a strong tendency, either implicit or explicit, to answer this question in the positive. (In fact, a lot of modern Western philosophy could be included, but we'll leave that out of consideration in what follows.)

Certainly not everyone would subscribe to the claim that it is logic that provides the best, let alone the only, way to deal with philosophical problems. But it seems clear that in a broader sense it is the *communis opinio* among analytic philosophers that philosophical problems require treatment on the basis of rational considerations, analyses and arguments.

Logical positivism, that established itself from the late 1920s onwards as an heir to the nineteenth century positivism of Comte, famously rejected speculative philosophical questions as non-scientific, and hence meaningless. Logic was the tool to expose them as such, and the tool that philosophers should use to embark on what was their true task: the analysis of the foundations of the sciences. All else would prove to be nonsense.[1]

That stark outlook turned out to be very influential. It was an important influence on philosophers such as Quine, Sellars, Davidson, Putnam and others, whose thought dominated philosophy in the US for decennia. Its emphasis on logic was formative there. That was not so much the case in the British school of 'ordinary language philosophy', which focussed, not, as the positivists did, on the construction of formal systems as philosophical tools, but rather on detailed descriptions of the way in which ordinary languages are used. But the rejection of speculative philosophy was a common thread.

What is important for the question at hand, is how analytic philosophy conceived of matters concerning areas such as ethics and aesthetics, since these would appear to be the primary candidates for philosophical matters that fall outside the realm of the rational (logical). Here we can observe that in analytic philosophy ethics and aesthetics are primarily treated at a meta-level. To put it bluntly (and thus slightly misleadingly), in analytic philosophy the topic of conversation no longer is how to live, or what art is, but how to think about such questions, what kind of theories one can develop about ethical and aesthetical value and how one should argue for those theories. That is to say, the actual development and propagation of ethical and aesthetical values have given reign to studying them from a meta-perspective. To be sure, the distinction between ethics and meta-ethics, or between aesthetics and meta-aesthetics, is not as absolute as the terminology might suggest. But it is indicative of the dominance of rational methods in analytic philosophy, and it also illustrates *how* that dominance plays out with respect to matters that we might perhaps regard as falling outside the realm of such methods.

[1] Thus, they promoted what they called a 'scientific view', among others by publishing a series of books under the title *Schriften zur wissenschaftlichen Weltauffassung* ('Writings on the Scientific View of the World').

What does that mean for the question at hand? One conclusion would be that in an important sense the answer depends on definition, of what counts as 'philosophical', and, of course, of what counts as 'rational'. Consider everyday reasoning. In many respects that often fails to satisfy the requirements we impose on reasoning in a scientific context. And if we align philosophy with science, everyday reasoning will not qualify as appropriate from that perspective either. But obviously, that does not mean that what we are concerned with in our everyday lives is not rational. On the contrary. It may be subject to different demands but reason plays a key role there as well.

So, from that perspective, rather than walling off philosophy from the everyday and seeking to establish it as a part of science, it would be better, it seems, to consider the borders between science and philosophy and those between philosophy and the everyday as permeable, allowing exchanges between the areas, movements across the border. That does not mean that there are no differences, but it does acknowledge mutual relevance.

Group 1:
"Can you think of atomic facts that are *not* 'facts of sense-perception? (Cf., OKEW, p. 43)"

First thing to note is that talk of 'atomic' and 'complex' comes with the assumption of a connection between the two: the atoms are what the complexes are constituted of. Thus, when one designates a certain

Much depends, of course, on what exactly one considers to be 'atomic facts'. This is a question that is not answered on the basis of empirical evidence, at least not in philosophy. Philosophers do not go out and investigate by observation and/or experiment, what counts as 'atomic': philosophy is not particle physics. But then what are the arguments that philosophy does consider?

Context plays an important part here. And it does so in two ways. First of all, there is the philosophical context in which we ask the question. If we operate in the context of epistemology (theory of knowledge) we probably use different criteria than when asking the same question in the context of ontology (metaphysics). The connection between atomic facts and sense perception could be established by definition. Of course, such a move does not come for free: it saddles one with the task of showing that all other facts that might be considered to be atomic, are not but can be accounted for in terms of these atomic facts of sense perception.

But there is also a second way in which context plays a role: what domain of facts are we considering? That's a broad question that we cannot answer here, of course. But consider the following examples.

One way of exploring the possibility of a negative answer is to look for facts where such a reduction to, or construction from, atomic facts of sense perception raises questions. Mathematics might be one source of such facts. Take arithmetic. Intuitively, there are very complex facts, such as Fermat's last theorem, but also very basic ones: 1 being smaller than 2, 1 plus 1 equalling 2, and so on. Can these atomic facts of arithmetic be reduced, directly or indirectly, to atomic facts about sense perception? The answer is not clear. For a mathematical realist the facts of mathematics are grounded in mathematical reality. That reality is not one that we perceive with the senses, that means that ontologically there cannot be any form of reduction. But what about epistemology? Here, arguably, there are links between sense perception and the

construction of basic arithmetical and geometrical notions.[2] Can these be considered reductive? That is a complex, issue, that is still open.[3]

Another interesting area is proprioception. Due to the presence of so-called 'proprioceptors' at various places in our bodies, we generally[4] know where our bodies are located in space-time, relative to other bodies, we know how we move our limbs, and so on. Knowing these facts is fundamental to our sense of self and our agency: without them we could not perform the most elementary actions and would not be able to locate ourselves in the environment we perceive. In that sense, proprioceptive acts are of a different nature than the basic forms of sense perception that Russell and others would refer to. We see perceive our surroundings, and the objects and people that are present in it. But we do not perceive ourselves in the same way: we do not see ourselves in a room in the same way that we see a table in there. This needs qualification, of course: we do look in the mirror to check our hairdo, but this sense perception of ourselves is only possible because of our proprioceptive abilities.

And then there is yet another reason to question the existence of atomic facts as a basis to which all facts can be reduced: Are there really independently motivated notions of atomicity and complexity, of things being simple or composed *as such*? Wittgenstein, in his *Philosophical Investigations* (1953), criticised his own positive answer to this question that he gave in his earlier *Tractatus Logico-Philosophicus* (1921), in which the concept of simplicity played a fundamental role:

> But what are the simple constituent parts of which reality is composed? – What are the simple constituent parts of a chair? – The pieces of wood from which it is assembled? Or the molecules, or the atoms? – "Simple" means: not composite. And here the point is: in what sense 'composite'? It makes no sense at all to speak absolutely of the 'simple parts of a chair'. (*PI*, section 47)

Contextually, yes, we can define what will count as complex and what as simple. But in an absolute sense? That is far from clear.

Group 2:
"Could there be such a thing as 'cognition without induction' that is *not* deductive? (Cf., OKEW, p. 27)

The idea that all knowledge is either deductive or inductive seems to fit a particular view on *scientific* knowledge. There are mathematics and logic (in a broad sense), in which theorems are derived from (self-evident) axioms in a strict, purely deductive manner. And there is empirical science, which is based on repeated observations and experiments and which derives conclusions by means of induction, i.e., by generalizing over a finite number of instances.

If we take our lead from science, it seems reasonable to assume that not just scientific knowledge but *all* knowledge belongs to either category. And this is apparently what Russell assumes. Everyday knowledge, then, is like scientific knowledge, but less rigorous, and hence more

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[2] See J. Stam, M. Stokhof, and M. van Lambalgen. Naturalising mathematics? A Wittgensteinian perspective. *Philosophies*, 7(4), 2022 for an overview of the cognitive science literature on this, and a Wittgensteinian take on this issue.
[3] For another perspective, see G. Lakoff and R. Núñez, *Where Mathematics Comes From*. Basic Books, 2000.
[4] 'Generally', because of such phenomena as the 'rubber hand illusion', a condition that leads to what is called 'proprioceptive drift'. See, e.g., Constantini M, Haggard P. The rubber hand illusion: Sensitivity and reference frame for body ownership. *Conscious Cognition*, 16, 2007.

defeasible. This seems to fit such cases as the knowledge we acquire in our primary education, where training and repetition play an important role. That may not be induction according to scientific standards, but it does proceed in phases and involves testing and learning from mistakes. .

But is all everyday knowledge like that? For example, what about the knowledge that goes into finding our way around campus, showing up at the appropriate classroom at the appropriate time for our logic class? It would be decidedly odd to say we learned that by induction. First of all, there is no generalisation involved. That 'Logic, Language and Philosophy' takes place in A309 on Mondays and Wednesdays from 3.20 pm to 4.55 pm is a specific fact, i.e., one that is not general by itself and that also does not lend itself to generalisation. And secondly and most importantly, learning it does not involve any multiplicity of observation or experiment.[5]

Group 3:
If, as Russell claims, 'logical knowledge is not derivable from experience alone', then what *is* its source? (Cf, p.30)

This is a key question within any epistemological framework that considers empirical knowledge, based on sensory experience, as the 'standard'. We do have mathematical knowledge, and logical knowledge, and unless we are prepared to subscribe to a radical form of empiricism and claim that those forms of knowledge, too, are just very, very general forms of empirical knowledge,[6] we need to account for its special status and for its source.

As was pointed out above, in the remarks on question 1, it seems appropriate to make a distinction between the ontological and the epistemological dimensions of questions such as these. One thing is to give an account of what mathematical or logical facts *are*, another thing is to do the same for what *knowledge* of such facts is, and how we acquire it.

As for the latter, it seems clear that experience plays a role in the acquisition of the basic skills involved.[7] And also acquisition of more advanced knowledge requires training and experience. In that respect there is no difference between knowledge acquisition in logic and mathematics and in empirical sciences. But in another respect, there is: justification. When we make a claim about empirical phenomena, we justify it by referring to what we have learned by observation and experiment. But if we justify a claim about a logical or mathematical matter, we do so by providing a proof. (Interestingly, this does *not* hold for basic facts, such as simple counting and measuring or basic inferences. What role justification plays here seems an altogether different matter. We will have a chance to say more about that later.)

That difference is often accounted for by pointing to an ontological difference between empirical facts and the facts of mathematics and logic. They are assumed to belong to different ontological spheres, which come with different ways of accessing what's inside those spheres. For mathematical realists this is an independent sphere, which we access by special means (often termed 'intuition'). For constructivists the mathematical realm is something that we construct using the mind, which means access is something we create. That, too, makes mathematics a realm that is distinct from the empirical world. For logic similar positions can be discerned.

[5] That we may have consulted the first semester teaching schedule more than once is an entirely different matter.
[6] As did, for example, John Stuart Mill, and, during a certain period, Russell himself as well.
[7] See the reference in footnote 2 for an overview.

Group 4:
"Frege likens the relation between natural and formal language to that between the eye and the microscope. (Cf., BS, preface, p. 6) What does that imply about their relationship? And how does that sit with the idea of 'reform'?"

One important thing to note about the relation between the eye and the microscope is that there is an asymmetry: the use of the microscope involves the use of the eye, but not the other way around. And that applies not just to the use, the eye is involved in the microscope's design, its construction, and so on. So, to think of the two as 'on equal footing' is confusing. The microscope is an *addition* to the perceptual abilities offered by the eye; it is *not* a stand-alone alternative.

Does that asymmetry transfer to the relation between natural and formal language? It seems so. In the same way as the use of the microscope depends on the eye, the use of a formal language depends on natural language. And the same holds for 'design & construction': formal languages are defined using natural language. To be sure, that may not always be a direct dependence: we can use a formal language to specify another one. But at the start of such a process we always find a natural language. And the same holds for the use of a formal language. In the end one does something with the results of using a formal language: evaluate them, put them to some use. And that involves natural language again. So, a formal language and the use that we make of it are intrinsically embedded in our natural language and the use that we make of that, and cannot really be separated from

That is not to say, of course, that using a formal language does not have advantages that using natural language for the same purpose does not. Obviously, we have developed formal languages for good reasons, and it is for such reasons (as well as reasons that we did *not* anticipate when developing them) that we use them. And the same goes for microscope. So, the idea of likening natural language and formal languages to the eye and a microscope makes a lot of sense.

But does it also make sense to see a formal language as a genuine alternative to natural language? In view of the embeddedness of formal languages in natural language that seems based on a misconstrual of their relationship. Sure, a formal language can do things that a natural language cannot: achieve a certain level of explicitness, preciseness, efficiency. But in the end, what a formal language does, and how it does it, when it does that is something that can be explained and understood ultimately only in natural language.

It is interesting to note that in current discussions about generative AI systems we encounter a similar (not the same!) situation. Even for their developers these systems function to a certain extent as a black box. They have set up the system, they know how to tune it and how, by doing so, influence its performance. But what goes on 'inside' them is not really survey-able. And to many that means that we do not really understand them.

But one may argue, all this applies *now*. However, humans evolve, among others by changing themselves. Increasingly, we are able to repair, or even replace, body parts by 'artificial' ones, we have extended our cognitive reach by all kinds of artificial means. So, why would language be exempt from this drive to enhance, or even replace?

With that question we enter even more speculative territory. For an answer really depends on how we construct human identity: what goes into making a human a human? It seems no exaggeration to state that, for as long as we know, language has been considered a necessary,

even a defining, characteristic of what it is to be human.[8] And 'language' here really means 'natural language'. Could that change? And if it did, what would be the consequences for the concept of human identity? From our current perspective would these creatures still count as human? From their perspective, would they still consider themselves human? No easy answers here …

[8] That only humans have language was something that for a long time was considered obvious. Recent research has shown that many other species have language, in some form or other. But acknowledging that does not by itself change the view that 'to be human is to have language'.